ASCII MASTER FLOWCHART

Tsunami and Coastal Hazard Management

12-panel revision companion | GS-III Disaster Management

ASCII MASTER FLOW — PANEL 1/12: Generation and hazard firewall

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EARTHQUAKE | LANDSLIDE | VOLCANIC ACTIVITY | METEORITE IMPACT
-> RAPID WATER DISPLACEMENT -> LONG TSUNAMI WAVES
TSUNAMI != TIDE (GRAVITY) != STORM SURGE (CYCLONE WIND/PRESSURE)
UNDERSEA EARTHQUAKE ALONE IS NOT SUFFICIENT: VERTICAL DISPLACEMENT MATTERS
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ASCII MASTER FLOW — PANEL 2/12: Propagation to coastal impact

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DEEP OCEAN -> FAST TRAVEL + LOW VISIBLE AMPLITUDE
SHALLOW WATER -> SLOWING + SHOALING + HEIGHT/CURRENT INCREASE
-> RUN-UP (VERTICAL) + INUNDATION (HORIZONTAL) + RETURN FLOW
FIRST WAVE NEED NOT BE LARGEST -> WAIT FOR AUTHORISED ALL-CLEAR
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ASCII MASTER FLOW — PANEL 3/12: Indian source geography and exposure

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EAST -> ANDAMAN-SUMATRA SUBDUCTION ZONE
WEST -> MAKRAN SUBDUCTION ZONE
26 DECEMBER 2004 -> CENTRAL INDIAN OCEAN DISASTER AND INSTITUTIONAL TURNING POINT
DOCUMENT-PERIOD FIGURES: >2,200 KM EXPOSED SHORELINE; 13 TSUNAMIS/300 YEARS, 3 IN ANDAMAN-NICOBAR
QUALIFY HISTORICAL FIGURES BEFORE USING THEM AS CURRENT COUNTS
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ASCII MASTER FLOW — PANEL 4/12: Detection-confirmation-advisory rail

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SEISMIC NETWORK -> SOURCE ANALYSIS -> TUNAMI-N2 SCENARIO
-> BOTTOM PRESSURE RECORDERS (OPEN OCEAN) + TIDE GAUGES (COAST)
-> OBSERVATION-INFORMED ITEWC BULLETIN / THREAT UPDATE
DETECTION != CONFIRMATION != ADVISORY != LOCAL EVACUATION
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ASCII MASTER FLOW — PANEL 5/12: ITEWC authority and timing

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ITEWC AT INCOIS, HYDERABAD -> OPERATIONAL SINCE 15 OCTOBER 2007
ABOUT 10 MINUTES -> EARTHQUAKE DETECTION
ABOUT 20 MINUTES -> ADVISORY DISSEMINATION
UNESCO-IOC IOTWMS -> REGIONAL TSUNAMI SERVICE FRAMEWORK
STATE / DISTRICT / LOCAL AUTHORITY -> EVACUATION AND ALL-CLEAR
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ASCII MASTER FLOW — PANEL 6/12: Modelling and bathymetry

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SOURCE SCENARIO + BATHYMETRY + TOPOGRAPHY + LAND COVER
-> TUNAMI-N2 -> TRAVEL TIME / RUN-UP / DEPTH / VELOCITY / INUNDATION
INDIAN NAVAL HYDROGRAPHIC DEPARTMENT -> BATHYMETRY AND 2004 RE-CHARTING
MODEL BOUNDARY = SCENARIO WITH UNCERTAINTY, NOT AN EXACT FUTURE LIMIT
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ASCII MASTER FLOW — PANEL 7/12: Evacuation decision

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STRONG/LONG SHAKING OR ABNORMAL SEA CHANGE -> SELF-EVACUATE
HORIZONTAL -> INLAND / HIGHER GROUND WHERE TIME ALLOWS
VERTICAL -> ENGINEERED, ACCESSIBLE, OFFICIALLY DESIGNATED REFUGE
MAP -> SIGNED ROUTE -> ASSEMBLY -> ACCOUNTABILITY -> DRILL -> REVISION
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ASCII MASTER FLOW — PANEL 8/12: CRZ and layered coastal protection

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CRZ NOTIFICATION 2019 -> CATEGORY-SPECIFIC CRZ-I TO CRZ-IV CONTROLS
ENGINEERED -> SHELTERS / RETROFITS / SEA WALLS / BREAKWATERS
ECOSYSTEM -> DUNES / MANGROVES / FORESTS / REEFS / WETLANDS / BIO-SHIELD
2004 EVIDENCE -> HEAVIER DAMAGE WHERE DUNES WERE MINED
NO UNIVERSAL SETBACK, SEA WALL OR ECOSYSTEM BUFFER ELIMINATES RISK
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ASCII MASTER FLOW — PANEL 9/12: Coastal economy and inclusive warning

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PORTS -> SHUTDOWN / HAZARDOUS CARGO / CHANNEL REOPENING
FISHERIES -> AT-SEA WARNING / BOATS / LANDING CENTRES / MARKETS
TOURISM -> MULTILINGUAL SIGNAGE / UNFAMILIAR ROUTES
ISLANDS + OLDER PERSONS + DISABILITY + PATIENTS -> ASSISTED EVACUATION
CONTINUITY PLAN MUST MATCH USER, SECTOR AND MOBILITY
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ASCII MASTER FLOW — PANEL 10/12: Tsunami Ready and readiness decay

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VENKATRAIPUR (GANJAM) + NOLIASAHI (JAGATSINGHPUR), ODISHA
-> INDIA'S FIRST UNESCO-IOC TSUNAMI READY COMMUNITIES, 7 AUGUST 2020
MAP -> EDUCATE -> WARN -> DRILL -> EVALUATE INDICATORS -> RENEW
COMMUNITY-LEVEL RENEWABLE RECOGNITION != NATIONAL CERTIFICATION
COUNTER MEMORY LOSS, STAFF CHANGE, DEGRADED SIGNAGE AND ROUTE CHANGE
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ASCII MASTER FLOW — PANEL 11/12: Response, recovery and named evidence

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EVACUATION -> SEARCH/RESCUE -> HEALTH/WATER/SANITATION/SHELTER
INDIAN NAVAL HYDROGRAPHIC DEPARTMENT: 7 SURVEY SHIPS AFTER 2004
-> AID + SEA-LANE REOPENING + SEABED RE-CHARTING
PORTS / FISHERIES / TOURISM / HOUSING / ECOSYSTEMS -> BUILD BACK BETTER
SAFER SITING MUST NOT DESTROY LIVELIHOOD ACCESS OR COASTAL RIGHTS
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ASCII MASTER FLOW — PANEL 12/12: UPSC answer spine and evidence limits

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DEFINE DISPLACEMENT -> TRACE PROPAGATION / SHOALING / CONSEQUENCE
-> MAP ITEWC / BPR / TIDE GAUGE / TUNAMI-N2 / LOCAL AUTHORITY
-> APPLY CRZ 2019 + LAYERED PROTECTION + ACCESSIBLE EVACUATION
-> USE 2004 + TSUNAMI READY EVIDENCE -> RECOVERY AND MAINTAINED READINESS
EVIDENCE LIMIT: ATTRIBUTE DATES, COUNTS, LEAD TIMES AND DAMAGE CLAIMS
OUTCOME TEST: A FAST ADVISORY IS AN INPUT; SAFE LAST-MILE ACTION IS THE RESULT
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